Paleomagnetism of Archean rocks from northwestern Ontario: Wabigoon gabbro, Wabigoon Subprovince

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The Wabigoon gabbro of the Archean Wabigoon greenstone belt in northwestern Ontario preserves a univectorial natural remanent magnetization (NRM) with $D = 246°$, $I = 12°$ ($k = 19.5$, $\alpha_{95} = 10.5°$, $N = 11$ sites). The precision is reduced if sample means are averaged, however ($k = 9.3$, $\alpha_{95} = 9.2°$, $N = 29$ samples). The paleomagnetic pole falls either at 160°W, 11°S ($\delta p = 5.3°$, $\delta m = 10.6°$), corresponding to an age of ~1300 Ma on the Laurentian apparent polar wander path, or the reverse of this, 20°E, 11° N, corresponding to a late Archean age (~2800 Ma). No ~1300 Ma igneous or metamorphic event is known in the area; a major west-northwest-trending dike about 9 km south of the gabbro yields a virtual geomagnetic pole at 122°W, 45°N and seems to be of Abitibi age (~2150 Ma) rather than Mackenzie age (~1250 Ma). A few gabbro samples and some greenstones from the intrusive baked zone have hybrid remanences in which a higher blocking temperature Kenoran-age (~2600 Ma) NRM is superimposed on the gabbro characteristic NRM. However, the Kenoran component may be a younger chemical remanent magnetization (CRM) residing in hematite. The hypothesis that the gabbro characteristic remanence is itself a hybrid of Kenoran and Keweenawan (~1100 Ma) NRM's, which would explain both the high between-sample scatter and the lack of a ~1300 Ma remagnetizing event, is considered but rejected because fewer than 10% of the gabbro samples exhibit multivectorial swings during alternating field or thermal cleaning. Two geomagnetic field reversals are recorded at interior sites, but only one or none is recorded near the margin of the intrusion. The different cooling histories of margin and interior, as well as the bulk of the other evidence, favour magnetization during initial cooling in late Archean time.

Le gabbro de Wabigoon de la ceinture archéenne de roches vertes de Wabigoon dans le nord-ouest de l'Ontario conserve une aimantation rémanente naturelle (ARN) univectorielle avec $D = 246°$, $I = 12°$ ($k = 19{,}5$, $\alpha_{95} = 10{,}5°$, $N = 11$ sites). La précision est moindre si on considère la moyenne des échantillons, toutefois ($k = 9{,}3$, $\alpha_{95} = 9{,}2°$, $N = 29$ échantillons). Le pôle paléomagnétique se situe soit à 160°O, 11°S ($\delta p = 5{,}3°$, $\delta m = 10{,}6°$), correspondant à un âge de ~1300 Ma déterminé avec la courbe de la dérive apparente des pôles laurentienne, ou la position inversée, 20°E, 11°N, correspondant à l'Archéen supérieur (~2800 Ma). On ne connaît pas d'évènements ignés ou métamorphiques dans cette région d'âge ~1300 Ma; un important dyke de direction ouest-nord-ouest localisé à environ 9 km au sud du gabbro fournit un pôle géomagnétique virtuel à 122°O, 45°N et semble d'âge Abitibi (~2150 Ma) plutôt que Mackenzie (~1250 Ma). Quelques échantillons de gabbro et de roches vertes provenant de la zone intrusive échauffée possèdent des rémanences hybrides pour lesquelles une température de blocage plus élevée de l'ARN d'âge kénorien (~2600 Ma) se superpose sur l'ARN caractéristique du gabbro. Cependant, la composante kénorienne peut correspondre à une aimantation rémanente chimique (ARC) plus jeune portée par l'hématite. L'hypothèse voulant que la rémanence caractéristique du gabbro soit elle-même une hybridation des ARN's kénorienne et keweenavienne (~1100 Ma), ce qui expliquerait la grande dispersion entre les échantillons et l'absence d'un évènement de réaimantation il y a ~1300 Ma, est envisagée mais rejetée car moins de 10% des échantillons de gabbro présentent des oscillations multivectorielles durant la désaimantation par champ alternatif ou par traitement thermique. Deux inversions de champ géomagnétique sont enregistrées à l'intérieur de l'intrusion mais à peine une ou aucune n'est enregistrée en bordure. Des épisodes différents de refroidissement de la marge et de l'intérieur de l'intrusion, auxquels s'ajoutent l'ensemble des autres observations, témoignent en faveur d'une aimantation durant le refroidissement initial à l'Archéen supérieur.

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Introduction

The "Superior Geotraverse" project was a cooperative investigation of the Archean greenstone and gneiss belts of the western Superior Province in northwestern Ontario, using a combination of geological, geochemical, and geophysical methods. A partial report of results appeared in 1979 (Canadian Journal of Earth Sciences, Vol. 16, No. 10), including a reconnaissance regional paleomagnetic study (Dunlop 1979*a*). The present paper is the first in a series reporting paleomagnetic and rock magnetic data for individual formations in and near the Superior Geotraverse region.

The Wabigoon and Shebandowan volcanic–plutonic belts in the southern part of the Superior Geotraverse area (Fig. 1*a*) have been regionally metamorphosed to greenschist facies. The regional grade of the intervening Quetico gneiss belt is lower amphibolite. The Wabigoon gabbro, the subject of this paper, is located in the Wabigoon belt just west of the Geotraverse corridor, near the village of Wabigoon, Ontario.

The regional geology of the Dryden–Wabigoon area is illustrated in Fig. 1*b*. Archean mafic metavolcanics and metasediments are intruded by a number of small mafic to ultramafic bodies, of which the Wabigoon

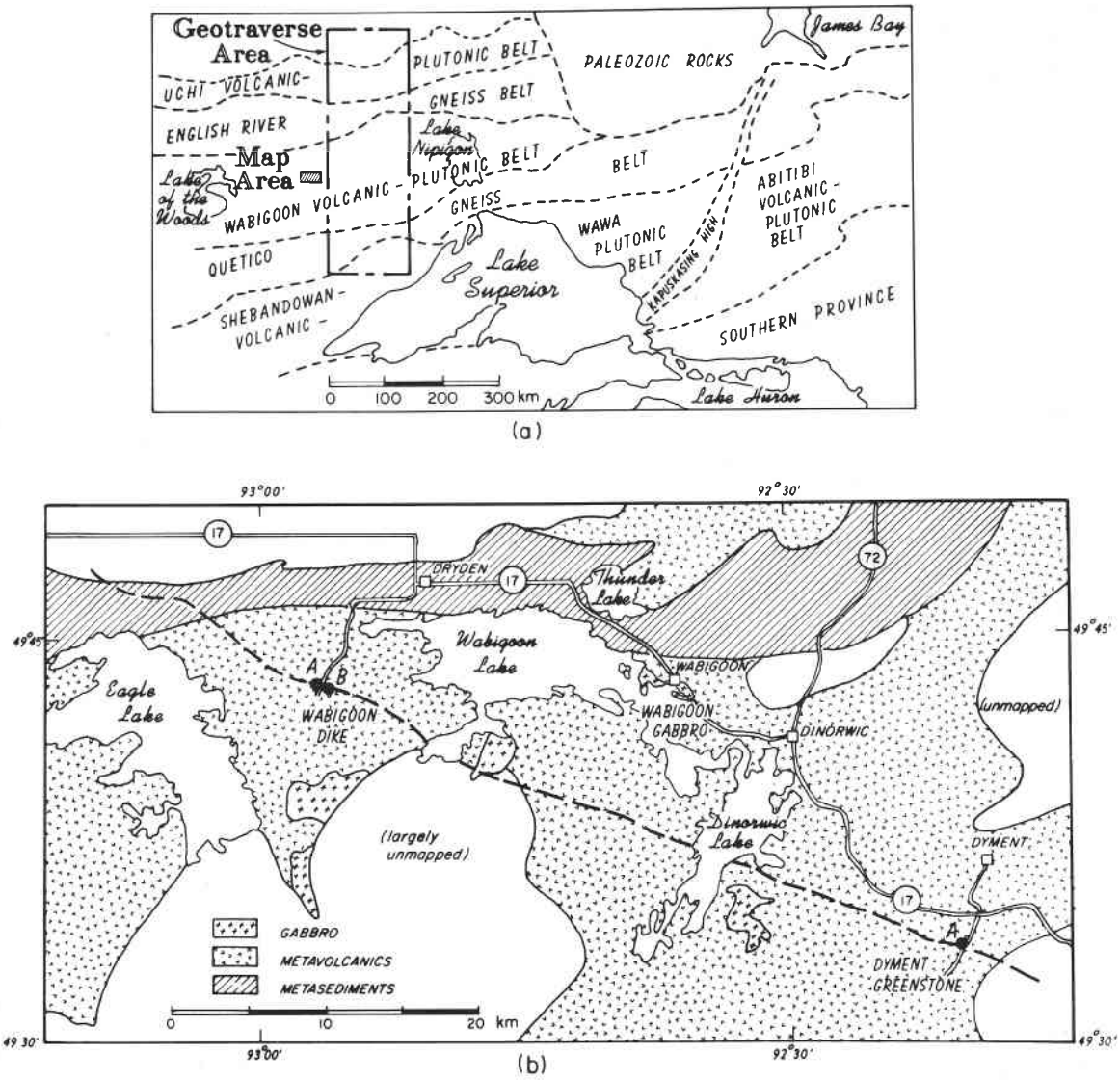


FIG. 1. (*a*) Subprovinces of the Superior Province in central and western Ontario, showing the locations of the Geotraverse area and the map area of (*b*). (*b*) Regional geology of the Dryden–Wabigoon area, indicating the location of the Wabigoon gabbro and sampling sites for the Wabigoon dike and the Dyment greenstone.

gabbro is one example, and by granitic to granodioritic stocks and plutons, probably of Kenoran age (~2600 Ma).

Dikes are relatively scarce in the western Superior Province, but a major west-northwest-trending quartz diabase dike crosses the map area just south of Wabigoon Lake. It may be continuous with other on-strike dikes to the east, paralleling Highway 17 from Ignace to Lac des Mille Lacs. Its strike length is at least 70 km, and may be as much as 270 km. Although it is mapped as Keweenawan (Satterly 1941), its age is an open question.

Sampling and measurement procedures

The Wabigoon dike was sampled at two sites (A and B in Fig. 1*b*) south of Dryden. South of Dyment, the dike could not be located in drift-covered terrain. Greenstones near the mapped contact were sampled at a single site (A in Fig. 1*b*). The Wabigoon gabbro was sampled at 11 sites (Fig. 2), and the greenstones it intrudes were sampled at two sites (site 2 about 0.25 km from the contact, and site 5 very near the contact). A total of 61 samples of the Wabigoon gabbro and its contact rocks and 16 samples of the Wabigoon dike and its contact rocks were collected. Three sites were field drilled; at the other sites, block samples were taken. The samples are not strongly magnetized and orientation by Brunton compass was sufficient.

Two specimens were cut from each drill core and at

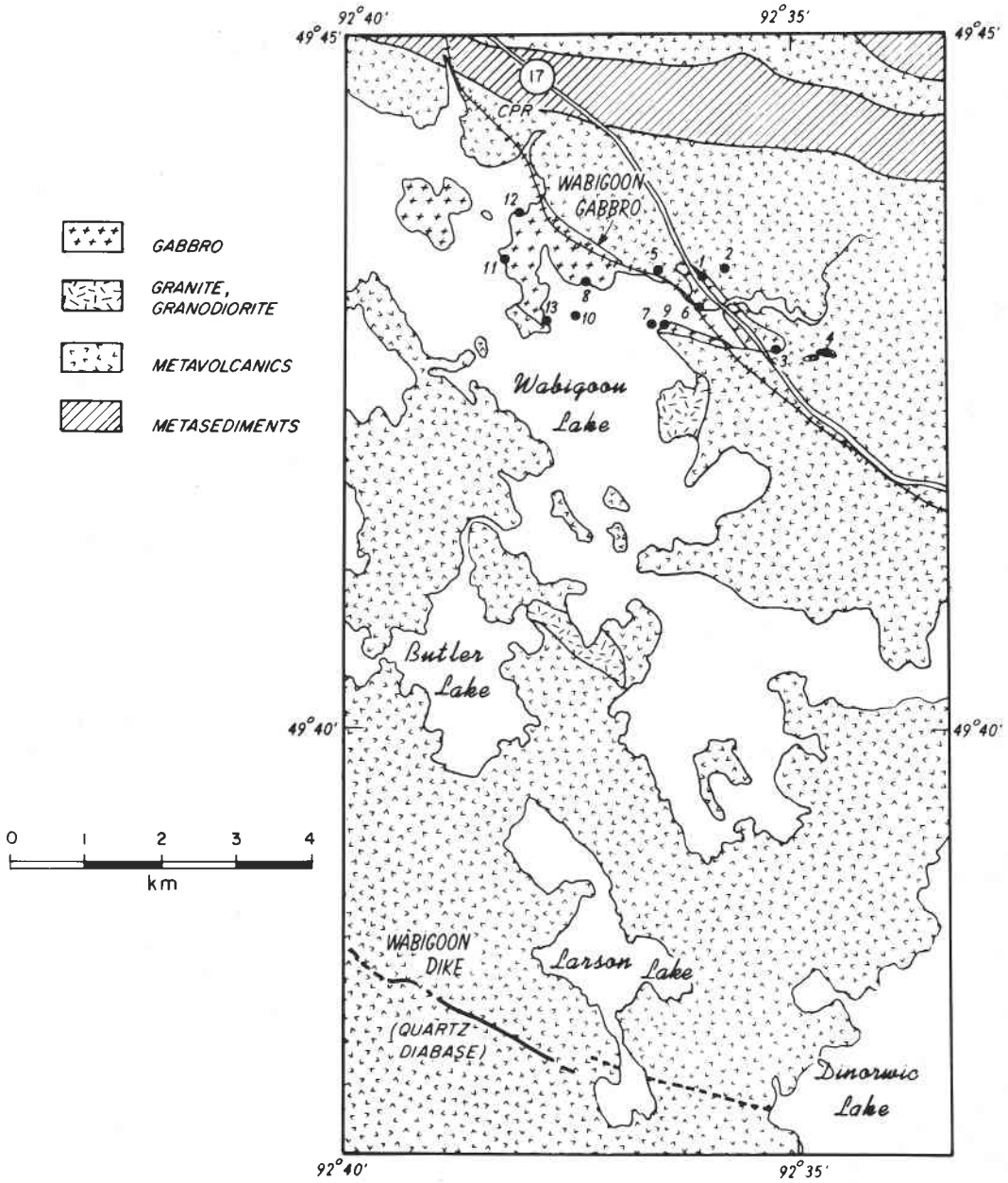


FIG. 2. Sampling sites in and near the Wabigoon gabbro.

least three specimens from each block sample. Whenever possible, two (sometimes three) specimens were alternating field (AF) demagnetized in steps to 1000 Oe (0.1 T), using a Schonstedt single-axis AF demagnetizer, and one specimen was thermally demagnetized in steps to at least 600°C (sometimes to 680°C). Early runs were made in a homebuilt furnace with field nulling by Helmholtz coils; later runs used a Schonstedt shielded furnace. A full program of measurements was carried out on each specimen. No blanket cleaning was used.

Natural remanent magnetization (NRM) measurements were made with a Digico spinner magnetometer. Thermomagnetic curves were measured in a vacuum of ~10 mTorr (~1.3 Pa) and a field of ~3 kOe (~0.3 T) with a recording Curie balance (Bailey 1980).

Results—Wabigoon gabbro

Thermomagnetic analysis

Strong-field, vacuum thermomagnetic curves, of which Fig. 3 is a particularly noise-free example, indicate that magnetite with a Curie temperature between 550 and 580°C is the principal magnetic phase. Minor

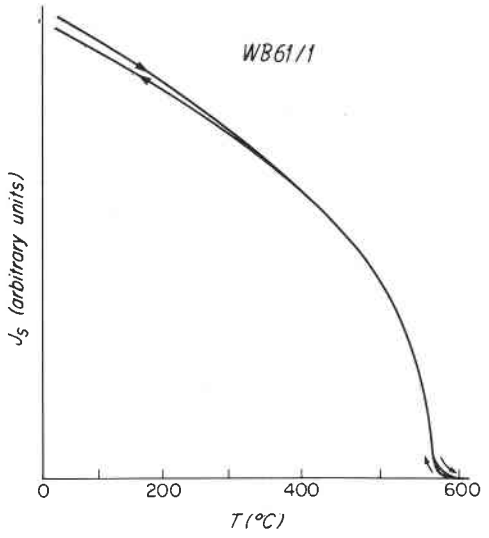


FIG. 3. Vacuum thermomagnetic curve for sample WB 61 (site 3), indicating nearly pure magnetite as the predominant magnetic mineral.

hematite and possibly also pyrrhotite or maghemite are revealed by thermal demagnetization.

AF demagnetization

Typical changes in the NRM vector during AF demagnetization are shown in the equal-area stereoplots of Fig. 4. A relatively soft NRM component approximately in the direction of the present Earth's field (PEF) is removed by 300–400 Oe (0.03–0.04 T) cleaning. In some cases, it is the predominant remanence (e.g., WB 1/2, 2/1, 2/2, 22/2); in others, it is a minor contaminant (e.g., WB 15/1, 16/1, 16/2, 46/1, 46/2, 46/3).

The characteristic remanence isolated by AF cleaning has south-southwest to west declination and intermediate positive to shallow negative inclination (reverse polarity, most specimens) or the reverse of this (normal polarity, e.g., WB 18/1, 18/2). The northeast direction is arbitrarily designated as normal because it is closest to the PEF.

Directional coherence is reasonably good except in rather weakly magnetized samples (NRM $\leq 10^{-6}$ emu cm^{-3} (10^{-3} A m^{-1})), which tend to exhibit a cyclic displacement of the magnetization vector at higher AF levels as the last axis of demagnetization is permuted. This "spruce-tree" effect is evident in WB 3/1, 3/2, 18/1, 18/2, 46/1, 46/2, and 52/1. It seems to be sample dependent and not related to instrumental anhysteretic remanences; it can be adequately corrected for by vector averaging *X*-last, *Y*-last, and *Z*-last demagnetization results at a given AF level (Zijderveld 1975).

Figure 4 shows a few examples of vector swings from the characteristic direction towards a south to south-southwest direction with shallow negative inclination (e.g., WB 15/1, 17/1, 19/2, 38/1, 38/2). This latter direction is well expressed in many units in the Geotraverse region (Dunlop 1979*a*) and elsewhere in the Superior Province (e.g., Schutts and Dunlop 1981). It corresponds to a Kenoran age of magnetization (~2600 Ma).

AF intensity decay curves for several samples with essentially single-component NRM's are given in Fig. 5. Samples like WB 3/2 that display prominent spruce trees (Fig. 4), have soft, rather erratic decay curves. Moderately hard characteristic remanences, with median destructive fields (MDF's) $\geq$400 Oe (0.04 T), are more typical of the collection (e.g., WB 18/2, 22/1). (The initial dip and recovery in the decay data for 18/2 are due to the elimination of a soft PEF remanence.)

Thermal demagnetization

Changes in the NRM vector during thermal demagnetization (Fig. 6) are generally similar to changes in AF cleaning. Initial PEF components are frequent (e.g., WB 2/4, 16/4) and, with the single exception of 2/4, have blocking temperatures (T_B's) $\leq$ 305°C, as expected for viscous overprinting at ordinary temperatures (Wilson and Smith 1968; Dunlop 1969).

The characteristic remanence is usually isolated between 305 and 575°C, but is occasionally stable to 625°C (e.g., WB 16/4, 26/4). Both polarities are found, normal (N) being less common than reversed (R). Hybrid or bivectorial remanences (N or R characteristic + Kenoran) are rare. In only two cases, WB 21/4 and 38/3, did the NRM vector stabilize at a southward negative-inclination direction. The NRM vectors of two contact rocks, WB 17/4 and 19/4, began to migrate southward at low temperatures, but the swings were short compared with the corresponding AF swings (WB 17/1, 19/2 in Fig. 4).

Magnetization reversals

Reversals of magnetization (usually R $\rightarrow$ N) in individual specimens were frequently observed during thermal cleaning. They were never observed in AF cleaning. Most reversals were close to 180° (e.g., WB 16/4, 21/4, 26/4 in Fig. 6), but a few were distinctly less than 180° (e.g., 19/4, 29/4). Of the 22 samples from sites 4–11 and 13 that behaved stably during thermal demagnetization, 11 underwent complete reversals and a further 8 began swings but dropped into the noise level before completing a reversal.

The observed reversals could have one of several causes: (1) reversal of the geomagnetic field during cooling (following intrusion or during later regional uplift); (2) self-reversal of thermoremanent magnetization (TRM) during cooling; or (3) overprinting of TRM by a reversed chemical remanence (CRM). The first possibility is the most likely, as the following discussion will show. If either field reversal or self-reversal occurred, the R and N remanences should occupy

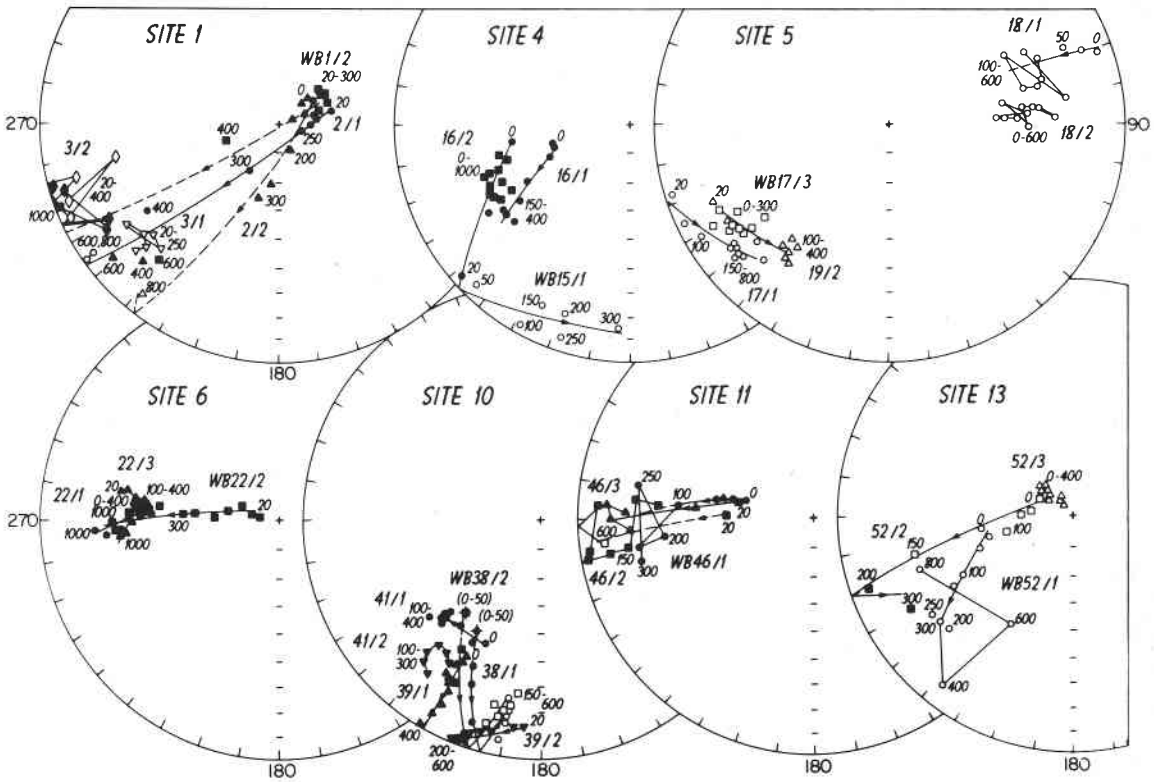


FIG. 4. Typical AF-cleaned vector NRM data for stably magnetized samples of the Wabigoon gabbro. Open points: upward vectors; solid points: downward vectors; labels on points are AF's in oersteds. Points marked with crosses and labelled with numbers in parentheses (e.g., WB 38/1, 38/2) are subtracted vectors over the AF range indicated. The plots are Lambert equal-area projections.

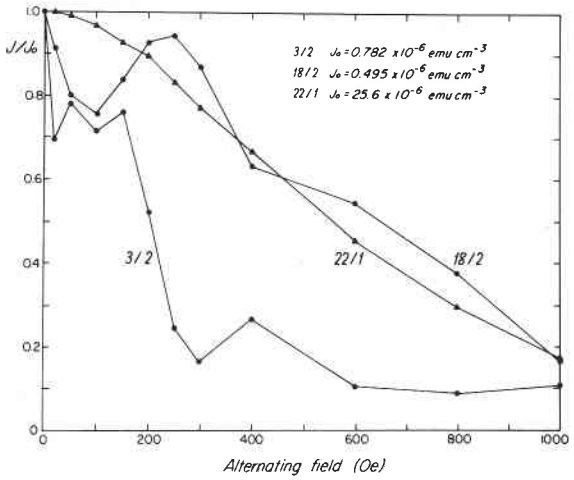


FIG. 5. Representative AF intensity decay curves for samples with approximately single-component behaviour.

non-overlapping parts of the T_B spectrum. CRM overprinting may overlap the T_B spectrum of the residual TRM, although it does not invariably do so.

Non-overlapping T_B spectra of R and N remanences are in fact typical of the Wabigoon collection. Figure 7 gives examples of Zijderveld (1967) plots, or Z-plots, in which matching straight-line segments in both horizontal-plane and vertical-plane projections represent separate NRM components whose T_B spectra do not overlap (Dunlop 1979*b*). The temperature at which linear segments representing R and N components intersect is the reversal temperature. It is quite sharp for all the specimens illustrated. (In the case of WB 2/4, thermoviscous overprinting of an R characteristic is recorded, rather than an R + N composite.)

The behaviour of WB 16/4 is representative of practically all gabbro samples taken at or near the margin of the intrusion (sites 4, 6, 9, 10, and 13). The NRM vector begins to migrate towards an N orientation above 350–400°C (Fig. 6; cf. also WB 21/4, site 6). However, the Z-plot shows that between 305 and 575°C only the R magnetization is being removed. The N magnetization is confined to the T_B range above 575°C and is presumably carried by hematite. Notice that the NRM vector of WB 21/4, after apparently stabilizing in a Kenoran direction between 500 and 560°C, likewise completes a reversal at 580–600°C (Fig. 6).

Although the R and N remanence in these samples occupy mutually exclusive T_B ranges, it is not possible

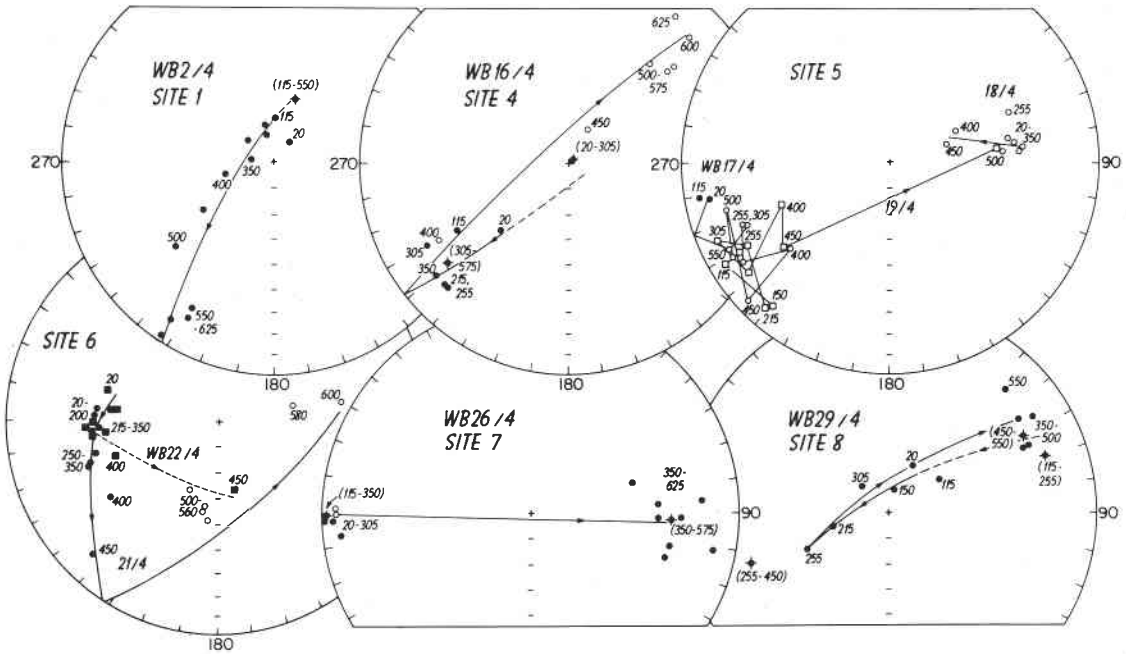


FIG. 6. Equal-area projections of thermally cleaned vector NRM data for the Wabigoon gabbro. Complete or partial reversals are recorded by WB 16/4, 19/4, 21/4, 22/4, 26/4, and 29/4. Symbols and labels as in Fig. 4, except that numbers here are temperatures (°C).

to rule out CRM overprinting, since the reversal temperature corresponds to the Curie temperature of magnetite. The N remanence could be a hematite CRM resulting from oxidation and would therefore be *younger* than the R magnetization. Note that this mechanism does not necessarily require a field reversal between the times that R and N were imprinted: CRM could have formed anti-parallel to the parent TRM.

The behaviour of WB 26/4 (site 7) is unique in the collection. Two successive R magnetizations of slightly different orientations (these were averaged in Fig. 6) reverse almost 180° to a smaller N magnetization with 350°C < T_B < 625°C (Fig. 7). A pyrrhotite–magnetite + hematite self-reversal or CRM–TRM combination is probable.

The best evidence that a field reversal occurred comes from samples of the interior of the intrusion (sites 8 and 11). WB 29/4 and 31/4 from site 8 undergo double NRM reversals (Figs. 6, 7): N (115°C < T_B < 255°C) → R (255°C < T_B < 450°C) → N (450°C < T_B < 600°C). WB 42/4 and 46/4 from site 11 record the complete N → R reversal, but the R → N swing is incomplete and obscured by noise.

What caused these double NRM reversals? A double self-reversal is improbable. TRM + two generations of CRM is not ruled out, but is unlikely because the reversal temperatures, 255 and 450°C, do not correspond to Curie points of any common magnetic minerals. Furthermore, thermal decay curves of NRM intensity for 29/4 and most other samples have no marked inflection near 300°C. Neither reversal can therefore be attributed, even indirectly, to pyrrhotite.

The likeliest sequence of events is the following. The interior of the Wabigoon gabbro cooled slowly enough to record two field reversals, at temperatures around 450 and 250°C. (Properly speaking, all laboratory T_B's should be scaled down, using the methods of Pullaiah *et al.* (1975) or Dodson and McClelland-Brown (1980), to allow for the effect of slow cooling.) The margin of the intrusion cooled rapidly and recorded only one field reversal (or possibly none, if the N remanence is a CRM).

The reversal record from the reheated contact rocks at site 5 is not simple. WB 17/4 and 18/4 have non-reversing NRM's, one of each polarity. WB 19/4 completes a reversal by 500°C, unlike gabbro samples at or near the contact whose swings are not complete until 575–580°C. Factors other than cooling history—perhaps physicochemical changes resulting from intrusion of the gabbro—must govern the reversal record at this site.

Averge directions

Sample-mean and site-mean magnetization directions are plotted in Fig. 8 for 29 stably magnetized samples from 11 sites. The characteristic remanence was isolated

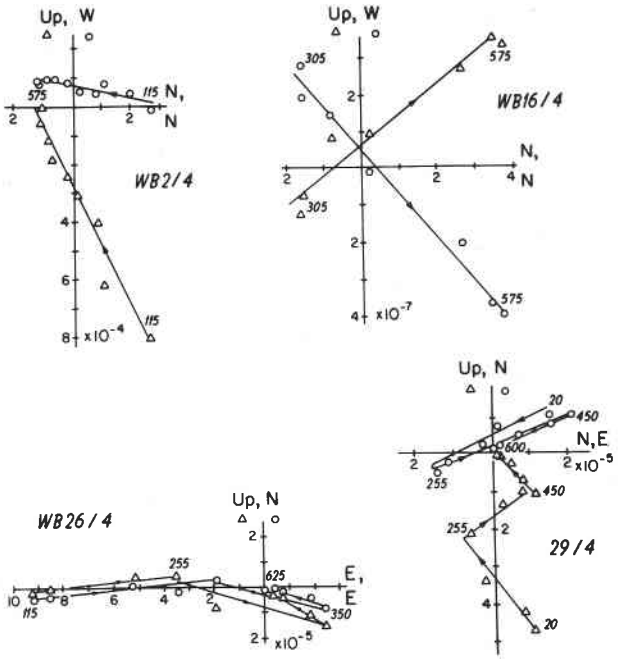


FIG. 7. Zijderveld vector diagrams of thermally cleaned magnetization data for several Wabigoon gabbro samples. In each case, triangles denote vector projections in the vertical plane (up vs. north or up vs. east) and circles are projections in the horizontal (north vs. east) plane. Labels are temperatures (°C). Linear segments indicate that a single vector is being erased over that temperature range. Subtracted vectors in Fig. 6 were calculated from linear segments in these and other similar vector diagrams.

by AF cleaning; thermally cleaned results were not used in the averaging. The other 26 samples were either weakly and unstably magnetized (NRM's usually $< 10^{-7}$ emu cm^{-3} (10^{-4} A m^{-1})), lightning struck (a particular problem with island and lakeshore sites), or so heavily overprinted by PEF remanence that the characteristic could not be isolated. Site 2 (greenstone 0.25 km from the contact) and site 12 had no stably magnetized samples, and sites 7 and 11 had only a single stable sample each.

The site 6, 7, and 11 means deviate markedly from the grand mean (Fig. 8*b*). However, these "deviant" directions are shared by a number of samples from other sites (Fig. 8*a*). The real problem is large dispersion among sample means at a given site, making it difficult to obtain a representative site mean if there are only one or two stable samples.

AF cleaned data are summarized in Table 1. The R directions corresponding to N results were used in averaging. The large within-site dispersion is reflected in rather low k values for site means and the low precision of the grand mean when sample data are averaged ($k = 9.3$) compared with the precision when site means are averaged ($k = 19.5$). Excluding data from sites 6, 7, and 11 improves the precision of the site-mean average (k increases from 19.5 to 47.2), as expected from Fig. 8*b*, but has an insignificant effect on the sample-mean average (k changes from 9.3 to 10.1). A k of about 10 best represents the scatter of the data.

However the averaging is done, the mean direction is close to $D = 240°$, $I = +10°$, and the corresponding paleopole falls near 160°W, 10°S. This pole position fits the North American Precambrian apparent polar wander path (APWP) (Irving 1979) at either ~2800 or ~1250 Ma. The former implies an Archean age of magnetization for the Wabigoon gabbro. The latter implies remagnetization at the time of the Mackenzie igneous episode, whose most obvious expression is the Mackenzie–Sudbury dike swarm to the north and the east of the Geotraverse area.

The next section investigates the possibility that the Wabigoon dike, whose strike is similar to that of the Sudbury dikes, is a member of this swarm, and is directly or indirectly responsible for the characteristic remanence of the Wabigoon gabbro.

Results—Wabigoon dike – Dyment greenstone

Wabigoon dike

Representative vector magnetization results during AF cleaning are depicted in Fig. 9 for sites A and B. Two characteristic directions are found. All specimens of samples A1, A3, A4, and A6 have a predominant steep, positive, west-directed magnetization, whose mean is $D = 267.0°$, $I = 79.8°$ ($k = 1997$, $\alpha_{95} = 2.0°$, $N = 4$ samples) with a virtual geomagnetic pole (VGP) at 121.5°W, 45.0°N ($\delta p = 3.8°$, $\delta m = 3.9°$). At site B, both reheated felsic country rocks (sample B1, whose 0–300 Oe (0–0.03 T) NRM perhaps represents the country-rock characteristic remanence) and altered quartz diabase dike samples (B2, B3, B4) swing towards this direction at AF's of 400–1000 Oe (0.04–0.1 T).

The VGP of this high-coercivity characteristic remanence resembles the paleopole of the Abitibi dikes (130°W, 41°N; Schutts and Dunlop (1981)), for which a primary magnetization age of 2150 Ma has been shown by a positive contact test and $^{40}Ar/^{39}Ar$ dating (Hanes and York 1979). It also resembles the primary paleopole of the 2160 Ma Nipissing diabase (103°W, 42°N; Roy and Lapointe (1976); plotted as pole ND7 by Irving (1979), Fig. 10) and Irving and McGlynn (1981, Fig. 23-10)).

The second characteristic remanence is a west-northwest or east-southeast shallow component exhibited by sample A8, taken from a 1 m wide, more felsic dike cutting the main dike at site A, and by samples A5 and A7, taken within 1–2 m of this subsidiary dike. To complicate matters, the NRM of specimen A1/2 (taken far from the small dike) swings to the second characteristic

TABLE 1. Wabigoon gabbro: sample- and site-mean magnetization directions after AF cleaning

Site	Sample	N	AF (Oe)[a]	D (°)	I (°)	Polarity	k	α_{95} (°)[b]
1	2	2	600	226	−8	R	52	
	3	2	100	238	−14	R	17	
	4	2	20	238[d]	2.5[d]	N, R	8.4	
3	9	3	50	223[d]	4.5[d]	N, R	29	23.3
	60	2	300	266.5	26	R	13	
4	14	2	250	225	2.5	R	50	
	15	1	20	228	2	R		
	16	3	300	236.5	35	R	140	10.4
	56	2	300	250	−7.5	R	32.5	
	58	3	200	270	−1	R	106	12.0
5	17	3	300	230	−22.5	R	81	13.8
	18	3	150	80	−35	N	231	8.1
	19	3	20	233	−8	R	265	7.5
6	21	2	150	251	48	R	40	
	22	3	1000	267	28	R	190	8.9
7	26	3	20	269.5	21	R	221	8.3
8	29	2	400	248[d]	−18[d]	N, R	30.5	
	30	2	50	42	−0.5	N	116	
	31	2	200	258.5	16	R	17	
9	33	2	200	258	23	R	15.3	
	34	2	0	219.5	−12	R	11.1	
	35	2	400	72	23	N	19.2	
10	38	2	[c]	216	49.5	R	189	
	39	2	800	241.5	−16	R	30.5	
	41	2	200	224	35	R	34	
11	46	3	150	263	22.5	R	103	12.1
13	52	2	300	235	−3	R	4.9	
	53	2	20	256	38.5	R	314	
	55	2	200	237	0.5	R	24	
1		3		234	−6.5		56	16.6
3		2		243.5	16		6.0	
4		5		241.5	6		11.1	23.9
5		3		240	1		5.9	56
6		2		260	38		24	
7		1		269.5	21			
8		3		242.5	−0.5		10.3	40.3
9		3		243	−4.5		6.7	52
10		3		228.5	24		5.1	61
11		1		263	22.5			
13		3		241.5	12		10.2	40.5
Mean, unit weight to samples		29		242.6	9.0		9.3	9.2
Paleopole				158.2W	13.7S			
				($\delta p = 4.6°$, $\delta m = 9.2°$)				
Mean, unit weight to sites		11		245.6	11.9		19.5	10.5
Paleopole				159.8W	10.7S			
				($\delta p = 5.3°$, $\delta m = 10.6°$)				
Means (sites 6, 7, 11 excluded)								
Unit weight to samples		25		239.5	5.4		10.1	9.5
Unit weight to sites		8		239.4	5.9		47.2	8.1

[a]Optimum cleaning field for minimum between-specimen dispersion.
[b]Not given if $N < 3$.
[c]Subtracted vector, 0–50 Oe (1 Oe = 0.1 mT).
[d]For mixed polarity samples, the R direction is given.

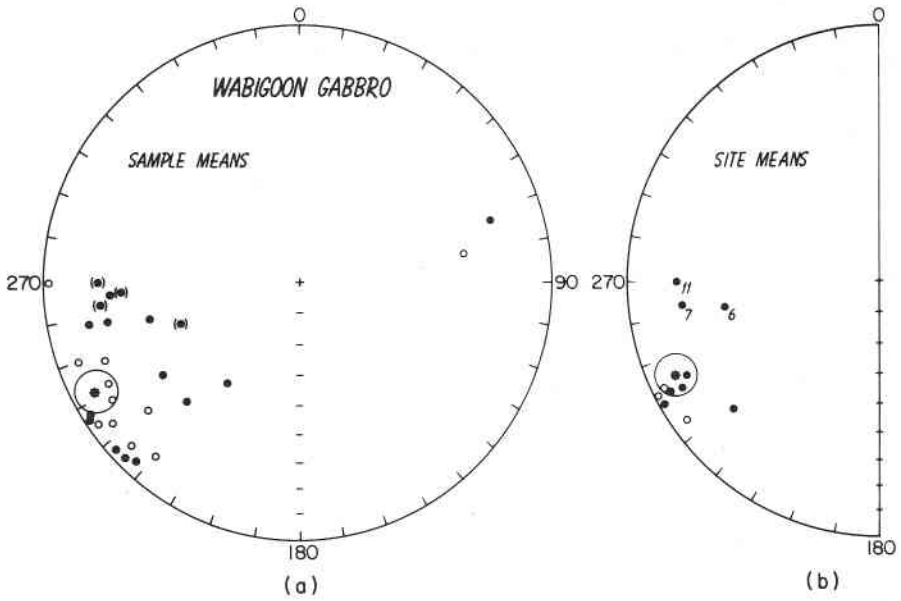


FIG. 8. Equal-area projections of AF-cleaned sample-mean and site-mean magnetizations for the Wabigoon gabbro. The grand mean, with its circle of 95% confidence, is shown as a star in either case. Sample means in parentheses are from sites 6, 7, and 11.

at 800–1000 Oe (0.08–0.1 T), but the felsic dike samples (e.g., A8/2) have NRM's that migrate towards the Abitibi-like characteristic above 150 Oe (0.015 T).

The directional grouping of samples A1/2, A7, and A8/2 is not 180° reversed to the grouping of A5. Neither group corresponds closely to any documented Precambrian age of magnetization younger than 2100 Ma and older than 1000 Ma, although the A5 grouping is perhaps consistent with latest Keweenawan (Halls and Pesonen 1982).

Detailed resampling will be needed to define either dike characteristic at an acceptable level of confidence. On the preliminary evidence, the dike was probably intruded and magnetized about 2100 Ma ago and was itself intruded by a minor felsic dike in late Proterozoic time (conceivably as late as Grenville time). The Wabigoon dike is apparently not of Mackenzie age, and likely played no role in the magnetization of the Wabigoon gabbro.

Dyment greenstone

All specimens of the five Dyment greenstone samples cluster about the mean direction, $D = 192.8°$, $I = -14.5°$ ($k = 183$, $\alpha_{95} = 5.6°$, $N = 5$ samples). The results are plotted in Fig. 10. The corresponding VGP falls at 69°E, 46.5°N ($\delta p = 2.9°$, $\delta m = 5.8°$), and is indistinguishable from paleopoles for the 2630 Ma Matachewan diabase (68°E, 50°N, Irving and Naldrett (1977); 70°E, 45°N, Schutts and Dunlop (1981)).

Site A is evidently too far from the contact to have been remagnetized, even partly, by the Wabigoon dike. It has, however, been totally remagnetized in latest Archean time, presumably as a result of the ~2600 Ma Kenoran Orogeny. It is worth noting in passing that the metavolcanics in the Dyment area deserve further detailed study, since they are almost unique among greenstones sampled throughout the Geotraverse area in possessing reasonably strong (3—20×10^{-6} emu cm^{-3} or $3–20 \times 10^{-3}$ A m^{-1}) and stable magnetizations.

Paleomagnetic poles

The paleopole for the Wabigoon gabbro (labelled WB) and VGP's for the Wabigoon dike (WD) and Dyment greenstone (DY) are compared with appropriate segments of the Precambrian APWP for North America in Fig. 11. Poles and labels in Fig. 11*a* follow Halls and Pesonen (1982, Fig. 4) for the most part. The remaining data follow Irving (1979, Figs. 9–11), with additions from the recent literature, as noted in the caption.

DY is indistinguishable from Matachewan diabase poles MD and MD2, and is also very similar to the ~2500 Ma pole PL of the Poohbah Lake complex (Dunlop 1979*a*) of the Quetico gneiss belt (cf. Fig. 1). WD is indistinguishable from the Abitibi diabase primary pole AD. However, WB is compatible with either an Archean age (not necessarily primary) of ~2800 Ma or a slightly pre-Mackenzie age of ~1300 Ma. Note that the oval of confidence for WB in Fig. 11*a* does not actually overlap the Mackenzie grouping CP, MX, SUD, and MA. WB is possibly consistent with an

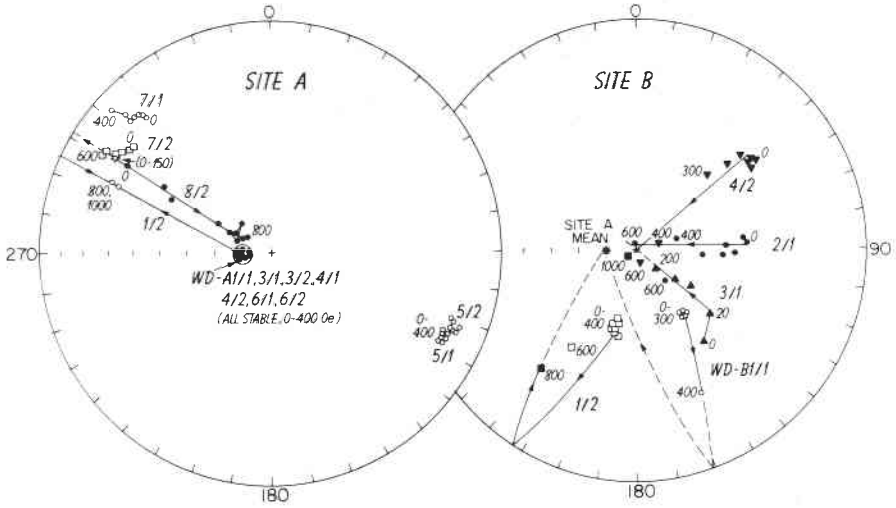


FIG. 9. AF-cleaned vector NRM data for specimens of the Wabigoon diabase dike. Symbols and labels as in Fig. 4.

earliest Keweenawan age, in fact, being intermediate between poles S1 and PM2, which have rather large confidence ovals.

Discussion

The age of pole WB is open to argument. Supporting an Archean age is the fact that generally similar poles (KK, AB, GR) that have been shown to be pre-Matachewan (Irving and Naldrett 1977; Schutts and Dunlop 1981) are widespread in the Abitibi volcanic subprovince to the east, whereas no ~1300 Ma poles are known from the Wabigoon or other volcanic belts of the Superior Province. However, the divergence between WB and the KK–AB–GR grouping in Fig. 11*b* is actually just as great as the difference between WB and Mackenzie-age poles like SUD and MA in Fig. 11*a*.

A more compelling argument against ~1300 Ma remagnetization is the lack of any indication of either Elsonian (~1400 Ma) or Mackenzie (~1250 Ma) age events in the Wabigoon area. To judge by its VGP (WD, Fig. 11*b*), the Wabigoon dike is of Abitibi (~2150 Ma) rather than Mackenzie age.

Structural arguments are inconclusive. The Wabigoon gabbro lies near the axis of an east–west syncline passing through Wabigoon Lake. No structural correction was made in arriving at pole WB. WB is therefore consistent with the ~2800 Ma Archean APWP only if magnetization postdated folding of the metavolcanics and metasediments. However, it is entirely possible that intrusion of the Wabigoon body postdated folding, or that the entire volcanic–plutonic sequence was remagnetized in the late Archean.

The fact that at least one geomagnetic field reversal was apparently recorded during cooling of the Wabigoon gabbro at first sight seems to support regional re-

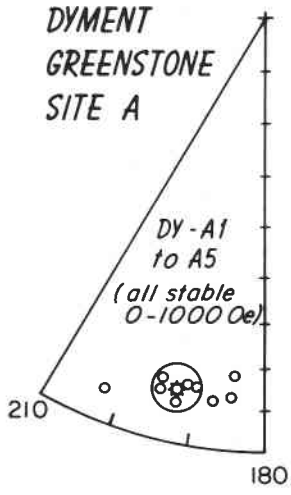


FIG. 10. AF-cleaned magnetization directions for specimens of the Dyment greenstone. All vectors are upward.

magnetization. Precambrian paleopoles tend to fall in groups of predominantly one polarity, implying rather infrequent reversals (Robertson and Fahrig 1971; Halls and Pesonen 1982). The probability of recording a reversal is therefore much greater if cooling was extremely slow, as it would be during regional uplift. However, the fact that interior sites record two reversals whereas marginal sites record at most one implies that the interior cooled slowly compared with the margin. This is the pattern expected for initial cooling.

Unfortunately, a direct contact test of the initial cooling hypothesis was unsuccessful. Volcanic rocks at site 5 were magnetized in the WB direction, but volcanics sampled away from the contact (site 2 and seven sites along Highways 17 and 72 in the Dinorwic–

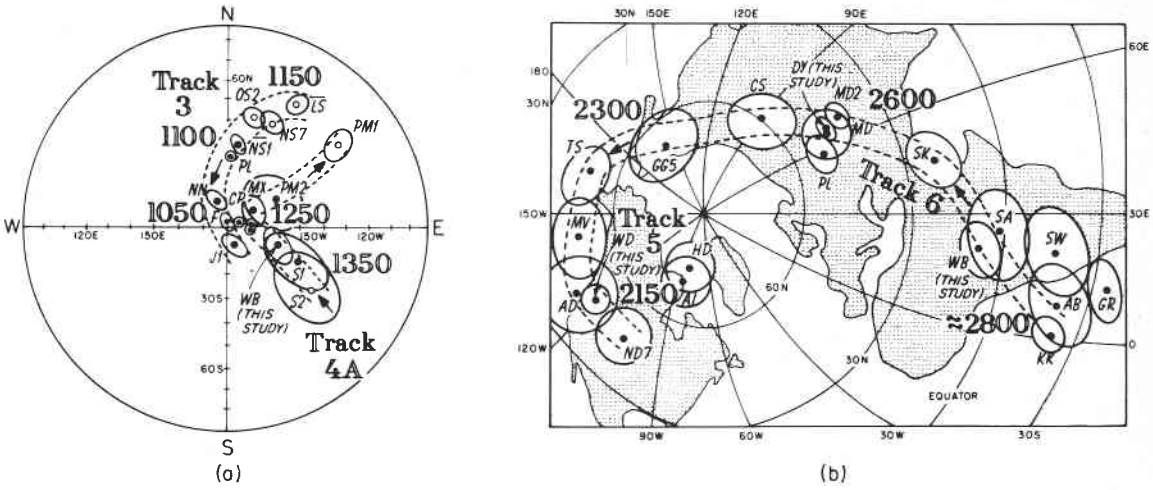


FIG. 11. (*a*) Late Proterozoic APWP for Laurentia, after Halls and Pesonen (1982, Fig. 4*a*), to which most of the labels are keyed, with the addition of Mackenzie-age (~1250 Ma) poles MX, CP, MA, and SUD from Irving (1979, Fig. 9) and pole WB of the present study. MA and SUD are unlabelled, but fall just south of MX and just east of CP and F. Solid points denote normal polarity (relative to this APWP) and open points denote reversed polarity. The position of WB on the APWP is consistent with an age of ~1300 Ma. (*b*) Archean and early Proterozoic APWP for Laurentia, after Irving (1979, Figs. 10, 11) and Irving and McGlynn (1981, Figs. 23–14), with the addition of poles AB, AD, and MD2 (Schutts and Dunlop 1981) for Abitibi belt country rocks, Abitibi dikes, and Matachewan dikes, respectively, PL (Poohbah Lake complex, Quetico belt, Dunlop 1979*a*), SK (Skead group volcanics, Abitibi belt, Ridler and Foster 1975), GR (Ghost Range complex, Abitibi belt, Geissman *et al.* 1982), and paleopole WB and VGP's WD and DY of the present study. The Wabigoon dike appears to be of Abitibi age (~2150 Ma); the Dyment greenstone was remagnetized in Kenoran times (~2600 Ma); and WB is consistent with a late Archean magnetization age of ~2800 Ma.

Wabigoon area) did not yield a stable NRM. Volcanics in the Dyment area did behave stably, but had been completely remagnetized (to a Kenoran direction) at the close of the Archean.

An alternative approach to dating WB is to determine its *relative* age compared with a superimposed NRM of known age. A few Wabigoon samples have a Kenoran-type (~2600 Ma) NRM component similar in direction to the Dyment overprint (Fig. 10 and VGP DY in Fig. 11*b*). Examples are WB 15/1, 17/1, 19/2, 38/1 and 38/2 (Fig. 4). In each case, the resultant NRM vector migrates from the WB direction towards a southerly, shallow, negative vector with progressive AF cleaning. The same pattern appears in thermal cleaning (Fig. 6: WB 17/4, 20–450°C; WB 19/4, 20–215°C), but the data tend to be noisier and the swings much shorter. Only WB 21–4 and 38/3 record complete swings that allow both vectors in the hybrid NRM to be resolved.

The simplest interpretation of these swings is that WB overprints (and is therefore younger than) the Kenoran remanence. Unfortunately this clearcut interpretation holds only if it can be shown that WB is a *thermal* overprint, i.e., a partial TRM. The test is the same one used earlier to distinguish NRM reversals involving only TRM from those due to CRM overprinting TRM, namely non-overlap of T_B spectra of the two NRM components. Unfortunately, the dividing temperature between WB and Kenoran NRM's in WB 38/3 is 560–580°C and 21/4 is 500–560°C. In neither case can one rule out the possibility that the Kenoran remanence is a high-coercivity, high-blocking-temperature hematite CRM that overprints the characteristic WB remanence.

Since there is no evidence for either ~1400 or ~1250 Ma igneous or metamorphic events in the Wabigoon area, is it possible that WB is an 1100–1050 Ma overprint that resulted from nearby Keweenawan rifting and volcanism? WB is very different from 1100–1050 Ma paleopoles (Fig. 11*a*). Suppose, however, that WB does not represent a simple univectorial remanence but an unresolved bivectorial sum of Keweenawan-age NRM and a small Kenoran vector. The bulk of the evidence is against such an idea. Only four samples (WB 15, 17, 19, and 38) show unmistakable swings toward a Kenoran direction, and two of these are reheated volcanics rather than gabbros. The NRM's of all other stable samples approach reasonably well defined WB end points and show no signs of migrating towards a southerly direction.

On the other hand, the distribution of sample-mean directions is broadly streaked between westerly, steeper, positive directions and southwesterly, very shallow directions. As Fig. 12 shows, this streaking follows a

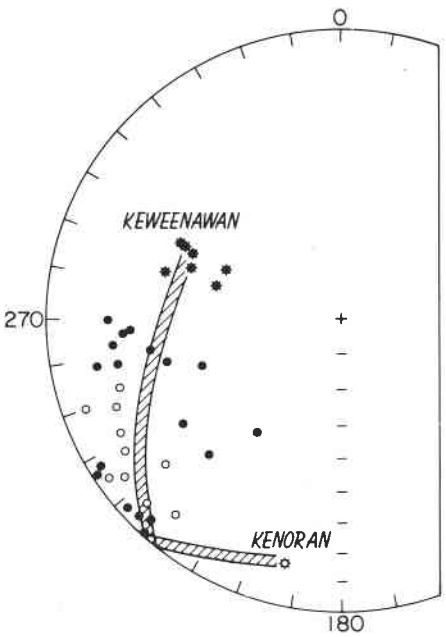


FIG. 12. The distribution of sample-mean Wabigoon gabbro magnetizations (Fig. 8*a*) compared with a great circle joining Keweenawan normal directions (Halls and Pesonen 1982, Table 2) and Kenoran directions (e.g., MD, PL, DY). The three normal-polarity results in Fig. 8*a* have been reversed in this figure. One possible explanation of the streaked distribution and high between-sample dispersion is unresolved bivectorial (Kenoran + Keweenawan) magnetizations.

great circle joining Kenoran and Keweenawan normal results. If WB actually represents a hybrid of Kenoran and Keweenawan remanences, in varying proportions, the streaked distribution and resulting high within-site scatter would be accounted for in a straightforward way. What is not obvious is how AF and thermal demagnetizations could fail, in more than 90% of the collection, to reveal the composite nature of the remanence. Such near-perfect coincidence of the coercivity and blocking-temperature spectra of the hypothesized Kenoran and Keweenawan remanences seems scarcely credible.

It must be concluded that most samples of the Wabigoon gabbro carry a univectorial remanence but are relatively poor and noisy paleofield recorders.

Conclusion

A Proterozoic age for the WB characteristic remanence is indicated by a straightforward interpretation of hybrid NRM's in the gabbro and its contact rocks. WB has lower T_B's and coercivities than the Kenoran component of NRM; it is therefore likely a younger overprint. It can also be argued that in view of the profound effect of the Kenoran Orogeny elsewhere in the Geotraverse region, it would be unusual if pre-Kenoran magnetizations were to have escaped Kenoran overprinting in all but a very few Wabigoon samples.

The evidence against a Proterozoic regional remagnetization is fairly compelling, however. There is no indication of a ~1300 Ma tectonic or heating event in the region. The reversal record of the Wabigoon body implies a chilled margin and slowly cooled interior. A contact test, which would have settled the question, failed because the unbaked country rocks are either unstably magnetized or totally reset by the Kenoran Orogeny.

On the basis of this evidence, it is tentatively concluded that the Wabigoon gabbro was intruded into already folded greenstones in late Archean time and has retained a characteristic remanence that largely escaped remagnetization in the Kenoran event. The geomagnetic field must have reversed fairly frequently at this time, in contrast to the situation during much of the Proterozoic, since two reversals were recorded during cooling of this rather small (1½ km × 5 km) body.

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